INT8 MatVec RTL optimization on VTR

Certified PPA estimate · open 45 nm FPGA proxy

A range-correct balanced adder tree improves the paired area-delay-power estimate by 8.3230%

Prepared by Göther Labs

Measurement completed: 16 August 2026

1. Executive result

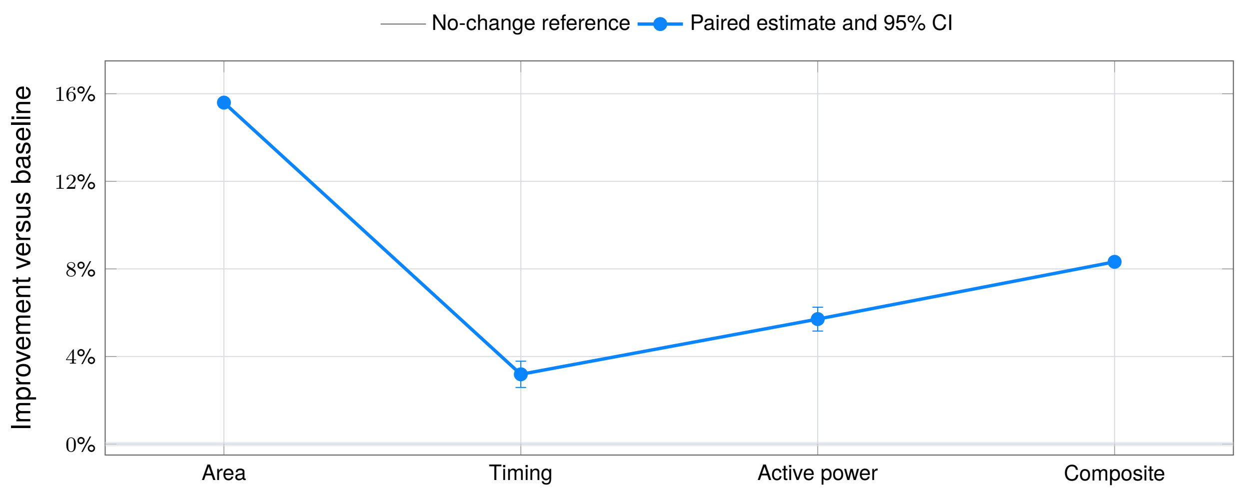
FROZEN BASELINE VERSUS OPTIMIZED RTL

The optimized RTL improves the equal-weight area-delay-active-power estimator by 8.3230% (95% CI: 8.1997% to 8.4462%) without changing the signed INT8 inputs, four signed INT32 outputs or exact matrix-vector arithmetic.

Primary metric	Baseline aggregate	Optimized aggregate	Paired improvement	95% CI / state
Total area	33,892,339 MWTA	28,607,064 MWTA	**15.5943%**	15.5355% to 15.6531% · **better**
Critical path	11.1642 ns	10.8085 ns	**3.1868%**	2.5843% to 3.7855% · **better**
Active total power	22.4567 mW	21.1749 mW	**5.7081%**	5.1622% to 6.2509% · **better**
Composite estimator	1.000000	0.916770	**8.3230%**	8.1997% to 8.4462% · **better**

Paired primary PPA improvement

64 fixed held-out certification pairs · lower is better · bars are two-sided 95% confidence intervals



The aggregate values are geometric means across the fixed held-out sample. The reported composite value 0.916770 is the paired geometric-mean estimator; the median of the 64 per-pair composite ratios is 0.916770. The estimator and its paired interval define acceptance.

2. Module and benchmark

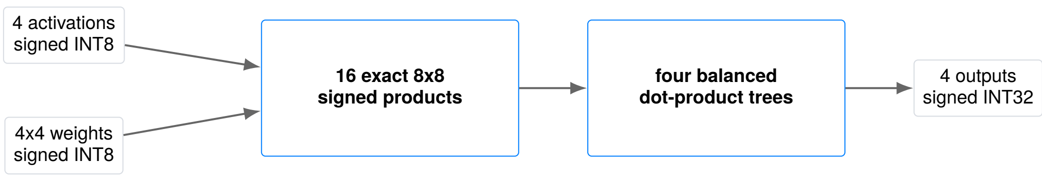
WHAT IS BEING OPTIMIZED

The evaluated module is an owned, Apache-2.0-licensed, synthesizable SystemVerilog implementation of a signed 4x4 matrix-vector product. Four signed 8-bit activations and sixteen signed 8-bit weights produce four exact signed 32-bit dot products:

$$y_i = \sum_{j=0}^{3} w_{ij} x_j, \qquad i \in \{0, 1, 2, 3\}.$$

Quantized linear, projection and convolution operators repeatedly use this arithmetic primitive. The case deliberately isolates the compute kernel; it is not a complete layer, memory system, neural network, accelerator platform or deployment.

```
module int8_matvec_4x4 (
  input  logic signed [7:0]  x0, x1, x2, x3,
  input  logic signed [7:0]  w00, w01, ... , w32, w33,
  output logic signed [31:0] y0, y1, y2, y3
);
```



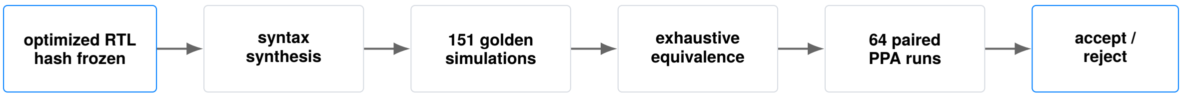
Only `int8_matvec_4x4.sv` is editable. Port names, signedness, widths, combinational behaviour and overflow semantics are frozen. Every result is compared with the checked-in owned baseline.

Artifact	SHA-256
Frozen baseline SystemVerilog	c2472728e2d52b29ad451f5c9779ebb749c318e9be048d03e6a6555c735eda53
Optimized SystemVerilog	25327189f78f238e909fa5940afff9e2b000547336cdc614ee9ea9c4633aa11f

3. Evaluation contract

FROZEN BEFORE CERTIFICATION

Dimension	Frozen choice
Editable artifact	`int8_matvec_4x4.sv` only; submissions frozen by SHA-256.
Functional behaviour	Exact ports, signed INT8 operands, signed INT32 outputs and combinational dot-product result.
Arithmetic semantics	Each 8x8 product is exact signed 16-bit; each four-term sum is exact and sign-extended to 32 bits.
Golden simulation	151 deterministic extreme, lane, row and seeded-random cases.
Formal semantics	Exhaustive two-state combinational equivalence for every 160-bit input assignment.
Physical flow	VTR/VPR commit `95f5c6de…`; pinned Linux/amd64 image; runtime network disabled.
Architecture	`k6_N10_I40_Fi6_L4_frac0_ff1_45nm.xml`.
Power	VTR PTM45, 0.9 V, $85\,^\circ$C; fixed active and idle activity profiles.
Search sample	5 paired placements; provisional ranking only.
Certification sample	64 disjoint held-out pairs; fixed sample with no extension.
Primary score	Equal-weight paired area, delay and active-total-power geometric mean.
Validity envelope	Fail closed on missing metrics, route failure, oversized multiplier structure or declared resource limit.



Exact decision rule. All correctness and evidence-integrity gates must pass. The composite one-sided 95% upper confidence bound must be below 1.0, and no primary metric may have a one-sided 95% lower bound above 1.0. Otherwise the result is inconclusive or regressed.

4. Verification before PPA

FAIL-CLOSED EVIDENCE FOR THE OPTIMIZED HASH

Gate	Published evidence	Result
Structural policy	Exact interface, patch and synthesizable SystemVerilog identities	**PASS**
Golden simulation	151 deterministic signed, extreme, lane, row and random cases	**PASS**
Combinational formal	All four outputs for every 160-bit binary input assignment	**PASS**
Physical validity	Width-sensitive mapping, resource envelope and complete metric extraction	**PASS**
Independent replay	Baseline and optimized certification records reproduce exactly	**PASS**

The mathematical reference evaluates signed integer dot products independently of the RTL expression shape. It covers zeros, unit impulses, alternating signs, -128, 127, maximum positive and negative accumulation, individual rows and lanes, plus deterministic seeded vectors. A mismatch stops evaluation before synthesis-based scoring.

The formal proof is unbounded because the design is combinational: it compares every output bit for every possible assignment of the twenty signed INT8 inputs. This is 2^{160} input assignments represented symbolically, not a claim that they were enumerated one by one. Inconclusive or timed-out proof is invalid.

Implementation validity rejects missing or non-finite metrics, failed routing, inconsistent area/frequency/power identities, multiplier hard blocks, multiplier operands above the declared width and excessive logic resources. The earlier syntax-level AST count is not an authoritative PPA metric.

```
python3 verify.py
```

> **Evidence boundary.** The public verifier establishes compact certificate consistency and exact published identities. The certificate records exact independent replay, but this INT8 package does not yet publish the raw VTR run trees required for a public raw-log provenance audit equivalent to the SHA-1 release asset.

5. Exact baseline-to-optimized diff

ALL CHANGED LINES FROM INT8_MATVEC_4X4.SV

The patch below is generated directly from the two checked-in RTL files. No testbench, reference model, architecture, activity profile, metric parser or tool flag is editable by the optimization candidate.

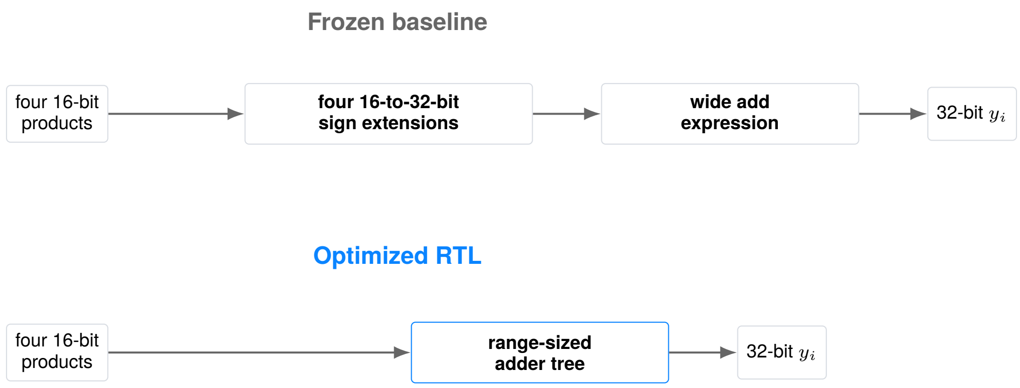
```
--- rtl/baseline/int8_matvec_4x4.sv
+++ rtl/optimized/int8_matvec_4x4.sv
@@ -32,10 +32,11 @@
     logic signed [15:0] p10, p11, p12, p13;
     logic signed [15:0] p20, p21, p22, p23;
     logic signed [15:0] p30, p31, p32, p33;
-    logic signed [31:0] e00, e01, e02, e03;
-    logic signed [31:0] e10, e11, e12, e13;
-    logic signed [31:0] e20, e21, e22, e23;
-    logic signed [31:0] e30, e31, e32, e33;
+    logic signed [16:0] s00, s01;
+    logic signed [16:0] s10, s11;
+    logic signed [16:0] s20, s21;
+    logic signed [16:0] s30, s31;
+    logic signed [17:0] d0, d1, d2, d3;

     always_comb begin
         p00 = w00 * x0;
@@ -55,26 +56,23 @@
         p32 = w32 * x2;
         p33 = w33 * x3;

-        e00 = {{16{p00[15]}}, p00};
-        e01 = {{16{p01[15]}}, p01};
-        e02 = {{16{p02[15]}}, p02};
-        e03 = {{16{p03[15]}}, p03};
-        e10 = {{16{p10[15]}}, p10};
-        e11 = {{16{p11[15]}}, p11};
-        e12 = {{16{p12[15]}}, p12};
-        e13 = {{16{p13[15]}}, p13};
-        e20 = {{16{p20[15]}}, p20};
-        e21 = {{16{p21[15]}}, p21};
-        e22 = {{16{p22[15]}}, p22};
-        e23 = {{16{p23[15]}}, p23};
-        e30 = {{16{p30[15]}}, p30};
-        e31 = {{16{p31[15]}}, p31};
-        e32 = {{16{p32[15]}}, p32};
-        e33 = {{16{p33[15]}}, p33};
+        s00 = {p00[15], p00} + {p01[15], p01};
+        s01 = {p02[15], p02} + {p03[15], p03};
+        s10 = {p10[15], p10} + {p11[15], p11};
+        s11 = {p12[15], p12} + {p13[15], p13};
+        s20 = {p20[15], p20} + {p21[15], p21};
+        s21 = {p22[15], p22} + {p23[15], p23};
+        s30 = {p30[15], p30} + {p31[15], p31};
+        s31 = {p32[15], p32} + {p33[15], p33};

-        y0 = e00 + e01 + e02 + e03;
-        y1 = e10 + e11 + e12 + e13;
-        y2 = e20 + e21 + e22 + e23;
-        y3 = e30 + e31 + e32 + e33;
+        d0 = {s00[16], s00} + {s01[16], s01};
+        d1 = {s10[16], s10} + {s11[16], s11};
+        d2 = {s20[16], s20} + {s21[16], s21};
+        d3 = {s30[16], s30} + {s31[16], s31};
+
+        y0 = {{14{d0[17]}}, d0};
+        y1 = {{14{d1[17]}}, d1};
+        y2 = {{14{d2[17]}}, d2};
+        y3 = {{14{d3[17]}}, d3};
     end
 endmodule
```

Attribution boundary. The measured PPA result belongs to the complete patch. The following pages explain three structural aspects of that patch; no exact fraction of the gain is assigned to one statement.

6. Rewrite 1: remove full-width extension bank

SOURCE: INT8_MATVEC_4X4.SV:35–43 AND 59–74

The baseline materializes sixteen signed 32-bit aliases before any addition. Every product is already exact in signed 16-bit form, so widening each term directly to the 32-bit output width is unnecessary.



For signed INT8 values $x, w \in [-128, 127]$, each product lies in

$$wx \in [-16{,}256, 16{,}384].$$

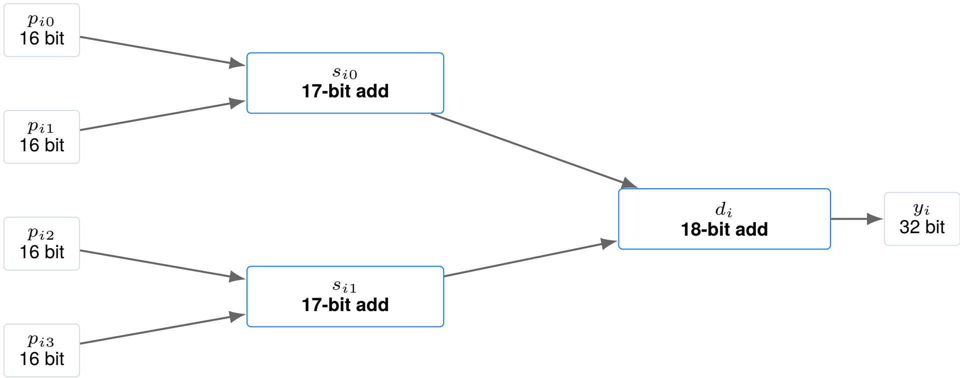
A pair sum therefore lies in $[-32{,}512, 32{,}768]$ and needs 17 signed bits. The complete four-term dot product lies in $[-65{,}024, 65{,}536]$ and needs 18 signed bits. No 32-bit intermediate is required before the interface boundary.

> **Formal result.** Exhaustive combinational equivalence proves that removing the extension bank preserves every output bit for every binary input assignment.

7. Rewrite 2: balanced range-correct accumulation

SOURCE: INT8_MATVEC_4X4.SV:59–70

The optimized RTL makes the intended adder topology and widths explicit. Each row first forms two independent 17-bit pair sums, then combines them in one 18-bit final sum.



For one row, the implementation is

$$\begin{aligned} s_{i0} &= \mathsf{sext}_{17}(p_{i0}) + \mathsf{sext}_{17}(p_{i1}), \\ s_{i1} &= \mathsf{sext}_{17}(p_{i2}) + \mathsf{sext}_{17}(p_{i3}), \\ d_i &= \mathsf{sext}_{18}(s_{i0}) + \mathsf{sext}_{18}(s_{i1}). \end{aligned}$$

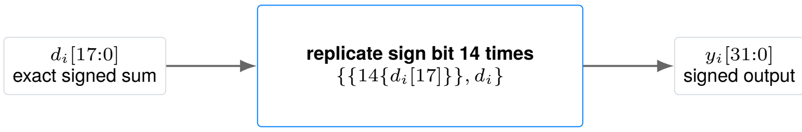
Integer addition is associative because every intermediate width covers its full mathematical range. The two-level topology avoids relying on tool interpretation of one wide chained expression and exposes width-sensitive mapping to synthesis.

Scope. This transformation is specific to four exact INT8 products and a 32-bit output contract. A different lane count, operand width, saturation rule or accumulation precision requires a new range proof and evaluation contract.

8. Rewrite 3: explicit signed output boundary

SOURCE: INT8_MATVEC_4X4.SV:72–75

The final 18-bit dot product is sign-extended once, at the frozen 32-bit interface. This makes the overflow convention explicit: the mathematical four-term INT8 sum always fits, so no truncation, wraparound or saturation occurs.



For every representable result d_i,

$$\text{value}_{32}(\{\{14\{d_i[17]\}\}, d_i\}) = \text{value}_{18}(d_i) = \sum_{j=0}^{3} w_{ij} x_j.$$

The output assignment changes neither combinational latency semantics nor the observable 32-bit value. It only prevents a wide internal representation from propagating backwards through the arithmetic cone.

> **Correctness result.** All 151 deterministic simulations and exhaustive formal equivalence agree on all four signed INT32 outputs. The accepted PPA measurement is therefore downstream of, and conditional on, exact arithmetic correctness.

9. PPA environment and statistics

OPEN 45 NM FPGA PROXY

The target matches the SHA-1 methodology case: VTR's homogeneous `k6_N10_I40_Fi6_L4_frac0_ff1_45nm` architecture, with six-input LUTs clustered ten per logic block and the architecture's routing and timing model. Power uses PTM45 properties at 0.9 V and $85\,^{\circ}$C. MWTA is minimum-width-transistor area, not square-micrometre physical area.

Two 2,048-cycle activity profiles are frozen: an active deterministic signed-INT8 workload containing extremes and a fixed xorshift64 sequence, and an idle all-zero data profile with the clock active. Active and idle power analyses reuse the same placement and route for a given design and blinded pair identity.

The primary per-pair score is

$$r_p = \sqrt[3]{r_{A,p}\, r_{D,p}\, r_{P,p}}, \qquad \widehat{r} = \exp\left(\frac{1}{64}\sum_{p\in P}\log r_p\right),$$

where A is total MWTA, D is post-route critical-path delay and P is active total power. Fmax is reported as the reciprocal of routed delay and is not a separate score term.

Two-sided 95% Student-t intervals use 63 degrees of freedom on paired log ratios. The one-sided bounds in the decision rule use the same predeclared fixed sample. These intervals quantify variation across the selected implementation pairs; they do not cover process, voltage, temperature, workload or tool-version uncertainty.

Pair policy. Search uses only 5 paired placements. Certification uses 64 disjoint held-out pairs after the optimized RTL hash is frozen. The public certificate replaces reserved placement-seed identities with stable pair labels; the optimizer and candidate do not see the certification pool.

10. Every paired certification result

PRIMARY PPA DISTRIBUTIONS

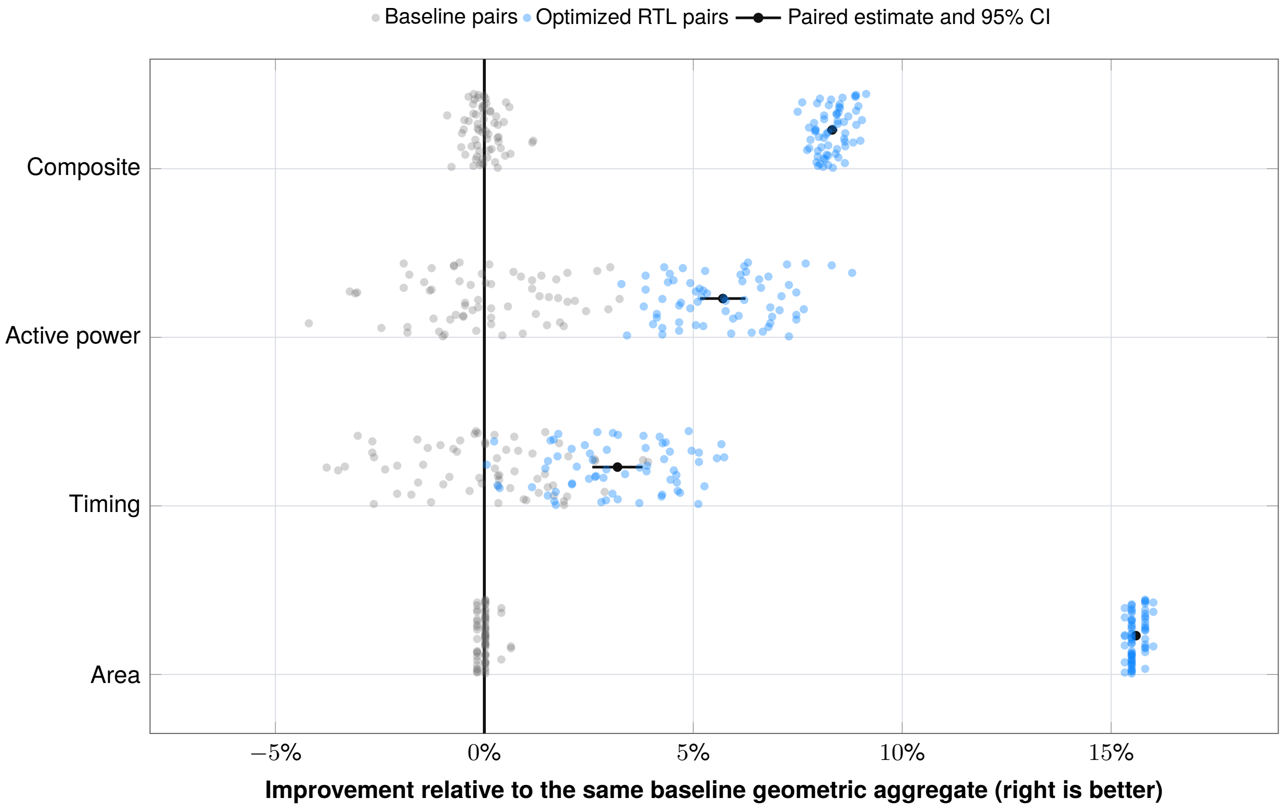


FIGURE 2 · OVERLAID BASELINE AND OPTIMIZED PAIR CLOUDS

Gray and blue points are absolute results normalized to the same baseline geometric aggregate. The same blinded pair receives the same vertical jitter. Black points and segments show the paired estimator and interval that determine confidence.

Metric	Wins	Ties	Losses
Area	64	0	0
Timing	54	0	10
Active power	64	0	0
Composite	64	0	0

11. Secondary metrics and evidence identity

MEASURED CONSEQUENCES AND COMPACT AUDIT TRAIL

Metric	Baseline aggregate	Optimized aggregate	Interpretation
Fmax	89.5717 MHz	92.5202 MHz	3.2918% increase; reciprocal of routed delay
Packed CLBs	368	304	17.3913% fewer in every pair
Logic elements	2586	2467	4.6017% fewer in every pair
Registers	72	72	Unchanged
Idle total power	9.1024 mW	8.1258 mW	10.7284% paired geometric improvement; excluded from score

The optimized design uses no multiplier hard blocks and remains within the frozen resource envelope. Dynamic and static components are retained per pair, but only active total power enters the primary score. None of these fields is inferred from an RTL node count.

Public evidence object	SHA-256 / identity
Compact certification record	e77ed39c79559af8cbbd2efeaf4fb67045740684ac4fc35699213c3cc1858906
Functional test-set identity	87abb71f1542981919fb62e299e0c19ff4cfacae20f41d50d0a41756d345c513
VTR architecture identity	262792caef81931ae09b77d252158bde03c78954175d4b761e6d437317575307
PTM technology identity	3080dea13bf7134109f8a83c79426ec0965e07639a61866fe9ae4bd05b5227cb
Optimized primary record	5f42601df61768be36b23594205c8d61d138bb4ed760443af0bbef0b43948a7d
Optimized replay record	441b9da134214162176d16a79e740dc50c1cdb1c9a244e6a1fa640b906304a6d

The certificate also binds distinct primary and replay records for the baseline. These hashes identify the evidence used to issue the compact record; they do not substitute for publishing the underlying raw files.

12. Reproduction and verifier trust model

WHAT EACH PUBLIC COMMAND ESTABLISHES

The repository separates three levels of evidence:

1. **Git identity**: checksums cover the exact compact package and the commit records its history.
2. **Compact consistency**: `python3 verify.py` validates strict schemas, RTL hashes, the exact patch, all 64 paired rows, derived metrics, formulas, confidence intervals, resource limits and acceptance rules.
3. **Independent replay attestation**: the compact certificate binds distinct primary and replay records and states that baseline and optimized records reproduce exactly. The raw VTR trees are not included in this public edition, so this level is not independently reparsed by the public verifier.

```
git clone https://github.com/juan-fernandez-gotherlabs/\
rtl-optimization-case-study.git
cd rtl-optimization-case-study/cases/int8-matvec

python3 verify.py
python3 -m unittest discover -s tests -v
```

The verifier uses explicit checks rather than Python `assert`, requires exact checksum coverage, rejects duplicate or unexpected schema fields, non-finite measurements, altered RTL, metric substitutions, score substitutions and any leaked held-out seed identity. Its adversarial unit suite exercises these cases in CI.

A clean physical rerun requires Linux/amd64, the pinned VTR commit, exact architecture and PTM files, fixed activity profiles and the sealed evaluation harness. This compact customer-facing package intentionally does not publish the optimization implementation or operational execution environment.

> **Trust boundary.** A checksum proves content identity, not authorship or raw-run provenance. The Git commit and future case release are publication anchors. Public raw-log audit parity with SHA-1 requires a separately reviewed, hash-addressed INT8 evidence archive.

13. Limits and customer transfer

WHAT THE RESULT DOES AND DOES NOT ESTABLISH

This report does not provide:

- ASIC synthesis, place-and-route or signoff;
- commercial-FPGA implementation data or Vivado results;
- extracted parasitics, PVT corners, clock-tree signoff, IR drop, EM, DRC/LVS, package, yield or production-test analysis;
- board, silicon, measured-power or measured-energy evidence;
- proof that the same structure improves a different matrix size, precision, architecture, technology or workload;
- exact PPA attribution to one line or one structural aspect of the patch;
- a complete neural-network layer, memory hierarchy or deployable AI accelerator;
- a public raw VTR evidence archive equivalent to the historical SHA-1 v1.3.1 asset.

The 64-pair interval models implementation-seed variability for the fixed VTR contract. It does not model silicon process variation or generalise automatically to a customer technology.

> **Transfer to a customer pilot.** Replace the academic proxy with customer-owned RTL, formal assumptions, libraries, constraints, activity, tools and signoff policy. The transferable capability is a small inspectable source change evaluated by a contract that can justify acceptance to the engineers responsible for the design.

Primary references

- VTR/VPR documentation
- VTR power estimation
- Yosys documentation
- Public case-study repository
- Göther Labs

> **Conclusion.** Within the declared VTR 45 nm FPGA proxy and exact signed-arithmetic contract, the optimized balanced-tree patch passes the published functional and formal gates and improves the paired primary PPA estimator by 8.3230% (95% CI 8.1997% to 8.4462%). The compact calculations and before/after RTL are publicly auditable; raw-run provenance remains outside this edition.